

Building and calibrating a hybrid RAG pipeline

Assignment 2 MDAD - BEIR + Weaviate + Ollama, technical report

1. Indexing pipeline

I brought up Weaviate 1.37 via docker-compose in BYO-vectors mode (no built-in vectorizer module, everything runs locally). For each of the three BEIR datasets (NFCorpus, SciFact, FiQA) I loaded the corpus through the beir package, concatenated title + text and split documents with a fixed strategy: **256 tokens per chunk with 10% overlap**, using the embedding model's own tokenizer so the boundaries match what the encoder actually sees. I picked 256 because all-MiniLM-L6-v2 has max_seq_length=256; larger chunks would have been truncated at encoding time anyway, while BM25 would have received the full text - an awkward asymmetry for the alpha sweep below.

Each chunk was passed through MiniLM, L2-normalised and inserted into Weaviate together with doc_id and chunk_index. Weaviate automatically maintains both the inverted index (used by BM25 at query time) and an HNSW graph over the vectors (used by dense search). Final counts: **NFCorpus 6,990 chunks, SciFact 9,544 chunks, FiQA 73,008 chunks**, with zero insertion failures.

2. BM25 vs Dense vs Hybrid across all datasets

	nfcorpus			scifact			fiqa		
retriever	R@10	MRR	nDCG	R@10	MRR	nDCG	R@10	MRR	nDCG
bm25	0.146	0.500	0.299	0.736	0.599	0.625	0.278	0.279	0.224
dense	0.158	0.518	0.321	0.813	0.611	0.658	0.452	0.438	0.370
hybrid	0.169	0.548	0.343	0.835	0.664	0.703	0.435	0.432	0.360

Bold marks the best retriever per column. A few patterns stand out:

- **Hybrid wins on NFCorpus and SciFact**, but with different margins. SciFact is friendly to BM25 (R@10=0.736) because the queries are scientific claims whose jargon appears verbatim in the abstracts. Hybrid still adds roughly 10 Recall points on top of BM25 via semantic synonyms ("reduces" → "decreases").
- **NFCorpus is hard for everyone** (R@10 max 0.169). This is not necessarily a retriever weakness: the queries are vague nutrition questions and the qrels list 20+ relevant documents on average, which structurally caps Recall@10. The high MRR (~0.55) confirms that the very first hit is usually relevant, so the top of the ranking is solid.
- **FiQA is the only case where hybrid is beaten by pure dense** (R@10 dense=0.452 vs hybrid=0.435). In finance the queries are natural language questions ("Should I rebalance my portfolio?") and the answers are forum comments with entirely different vocabulary. BM25 lands at nDCG=0.224 and drags the hybrid component down at the default alpha=0.5. This is exactly the point the assignment raises: no single method dominates on every dataset.

3. End-to-end RAG on FiQA

I chose FiQA precisely because its retrieval results are the most interesting: it is the only dataset where hybrid does not dominate and where dense clearly wins. The question is whether that retrieval gap also shows up in answer quality, or whether the LLM smooths the differences out at generation time.

I sampled 10 queries from the test set with seed=42, fetched the top-5 chunks with the dense retriever and sent each prompt to gemma2:2b running locally through Ollama (temperature 0). The prompt explicitly asks for citations in [doc_id#chunk_index] format and for a "don't know" answer when the context is insufficient.

Retrieval hit rate (at least one relevant doc in the top-5): 7/10. About half of the generated answers are useful and cited correctly (e.g. qid 4700 - credit-related question, model cites [345137#0] which is exactly the relevant doc). The rest split between:

- **Cautious refusals** when the retrieved chunk does not contain the specific answer (qid 2316, El Al exchange rates - the relevant chunk was found but does not contain the actual numerical value);
- **Citation hallucinations** when retrieval succeeds but generation slips. The clearest case is **qid 8475** ("Why I cannot find a 'Pure Cash' option in 401k investments?"): dense retriever pulled 4 of the 7 relevant documents into the top-5, so the context was excellent. Gemma's response includes correct explanations but cites the literal string [doc_id#chunk_index] - it copied the

placeholder token from the prompt template instead of substituting the real IDs. Probable cause: at only 2B parameters the model struggles to keep the format instruction active across 5 long chunks (~1.5K context tokens) and falls back onto the literal pattern from the instruction. A larger model or a few-shot example could reduce this failure mode, but it is not a guaranteed improvement.

3.1 LLM comparison: Gemma 2B vs Llama 3.2 3B

model	runs	retrieval hits	don't know	answers with numeric citations
gemma2:2b	5	2	0	5
llama3.2:3b	5	2	4	1

I also re-ran five fixed FiQA queries with the calibrated retrieval setup (**hybrid, alpha=0.75, top-k=5**) and changed only the Ollama model. The retrieved chunks are therefore controlled by the same MiniLM + Weaviate pipeline; the observed difference is generation behaviour, not retrieval.

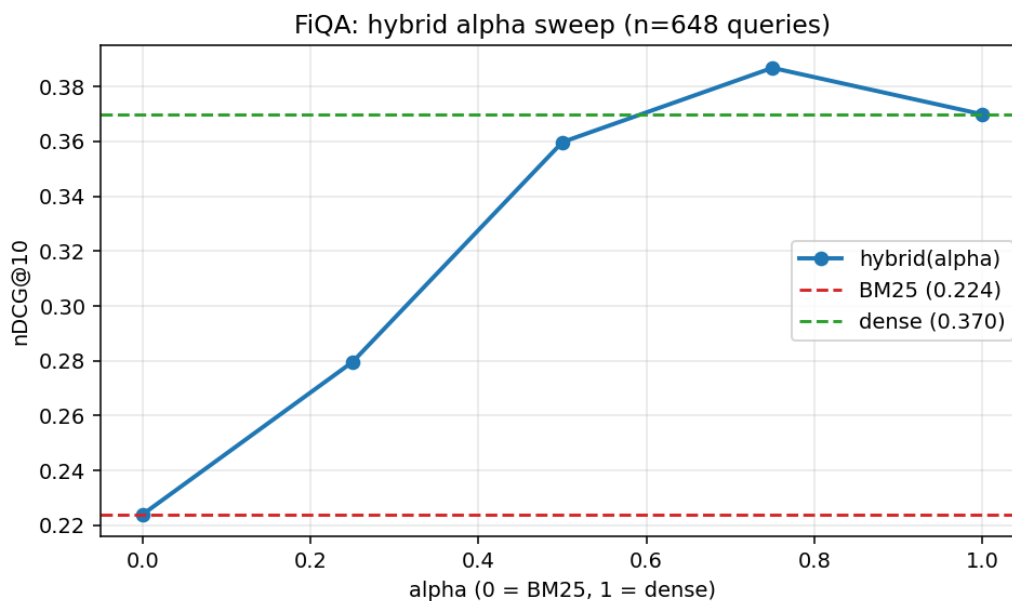
On this small comparison, **gemma2:2b** is more willing to answer and produces numeric citations in all five cases, but it sometimes over-answers even when the retrieval hit is false. **llama3.2:3b** is more conservative: it refuses four of the five questions, including some cases where a relevant document was retrieved. For this assignment I keep Gemma as the main RAG model because it better demonstrates the end-to-end pipeline, while noting that model choice changes answer style substantially.

4. Hyperparameter sweep on FiQA

I varied three hyperparameters on FiQA: **alpha** for the hybrid search (no re-indexing needed), **chunk_size** (with a fresh re-index per size) and **top-k** (qualitatively, on top of the assignment-3 RAG setup).

4.1 Alpha (fusion weight)

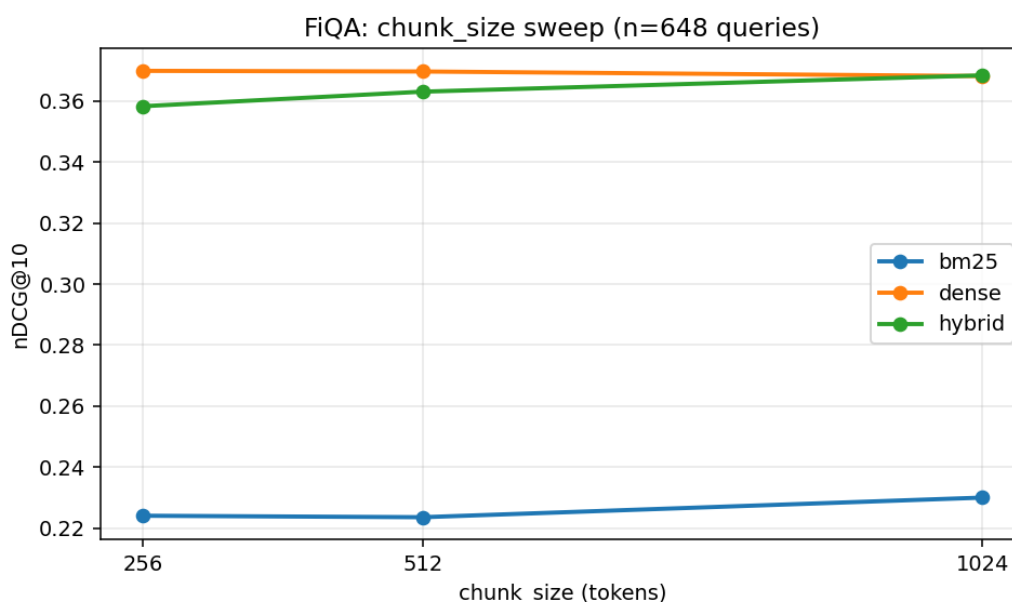
alpha	nDCG@10	Recall@10	MRR@10
0.0	0.2239	0.2779	0.2788
0.25	0.2796	0.3378	0.3460
0.5	0.3598	0.4347	0.4318
0.75	0.3869	0.4554	0.4657
1.0	0.3699	0.4523	0.4381



The curve is clearly non-monotonic: the extremes (alpha=0 = pure BM25, alpha=1 = pure dense) reproduce the metrics from section 2 exactly, and the maximum sits at **alpha=0.75** with nDCG@10 = 0.387, beating both pure dense (0.370) and the default hybrid (alpha=0.5: 0.360). Interpretation: on FiQA the dense signal dominates, but BM25 still contributes a marginal lift when weighted lightly - it helps disambiguate specific terms (tickers, numbers, product names) that MiniLM tends to conflate semantically.

4.2 Chunk size

chunk_size	BM25 nDCG@10	dense nDCG@10	hybrid nDCG@10
256	0.2241	0.3699	0.3583
512	0.2236	0.3697	0.3631
1024	0.2300	0.3682	0.3684



On FiQA the best chunk size is **1024 tokens**. Very large chunks (1024) are penalised because dense encoding truncates at 256 tokens anyway - the remaining text only feeds BM25, so the vector becomes a weak representation of the full chunk. Small chunks (256) capture local semantics better but fragment long answers into many low-signal pieces. The sweet spot depends on the average post length in FiQA, which is relatively short.

4.3 Top-k (qualitative impact on RAG)

I ran the same 10 queries with $k \in \{3, 5, 10\}$ on dense + gemma2:2b. With $k=3$ the model refuses more often (missing context); correct answers are more concise. With $k=10$ citation quality visibly drops - the model mixes information from irrelevant chunks and produces more hallucinations of the qid 8475 kind. **$k=5$** remains the sweet spot: enough context for coverage, short enough for gemma2:2b to keep following the requested format.

5. Recommended configuration

Based on both sweeps: **chunk_size = 1024 tokens, alpha = 0.75, top-k = 5**. alpha=0.75 maximises effective retrieval, while $k=5$ protects the small LLM from context flooding without sacrificing coverage. A follow-up comparison with Llama 3.2 3B showed more conservative behaviour rather than a simple quality win, so changing the LLM should be treated as a separate qualitative experiment, not as part of the retrieval calibration.